Daniel Navarro

Data Scientist & AI Engineer | RAG/LLM Systems, Data Engineering & BI | MA Public Policy (CEU)

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danielnavarro.work | Geneva, Switzerland (open to relocation)

I build data and AI systems end to end, from scraping and ETL to **RAG pipelines**, models and the dashboards people decide with. I combine applied **machine learning** (tree-based, time series, clustering) and modern **AI** (RAG, vector and graph databases, local LLMs) with production-minded pipelines in **Python, SQL, Docker and AWS/Azure**, for private-sector clients and for the UN in Geneva. My training in anthropology and quantitative policy analysis means I build for the people who use the output, and explain it clearly to technical and non-technical audiences.

CORE COMPETENCIES

Applied AI & RAG: RAG pipelines, vector and graph databases, embeddings, knowledge graphs, semantic search, LLMs (Hugging Face, local models via Docker Model Runner), NLP, automated report generation; agentic coding with **Claude Code**.

Machine Learning & Statistics: Python (pandas, scikit-learn), R; supervised and unsupervised ML (tree-based, regression, clustering), time series (ARIMA, XGBoost, NARX), deep learning, topic models and transformer sentiment.

Data Engineering & Cloud: ETL pipelines, web scraping, REST APIs, n8n automation, data architecture (ERDs, schemas, user flows); SQL (PostgreSQL, SQLite), Docker, AWS, Azure, Git.

BI & Visualisation: Power BI, Preset, Dash, Plotly, custom CRMs, KPI reporting and data storytelling; QGIS, GeoJSON, Copernicus/Sentinel satellite data.

Quantitative Policy & Research: impact evaluation and causal inference (TWFE, event studies), survey-weighted regression, procurement and corruption-risk analytics, MEAL.

PROFESSIONAL EXPERIENCE

Instrumental Partners LLC — **Geneva (remote)**
Data Engineer & BI Consultant — *Jul 2025–Present*

- Built an **AI RAG and graph knowledge base** with automated competitor and market-intelligence reporting over **1M+ federal procurement contracts** and 8k+ competitors, using proprietary RAG pipelines and local LLMs.
- Engineered Python web-scraping, API and **ETL pipelines** structuring procurement data (FPDS, DIBBS), and designed the data architecture from ERDs and schemas to user flows.
- Applied ML and statistical models for **pricing and bidding recommendations** in a custom CRM, with Apollo API enrichment for prospecting workflows.
- Built the **BI dashboards** clients run on (Preset, Power BI, SQLite), and shipped the firm's website end to end (content, case studies, SEO) with n8n content and news-monitoring automation.

UN OCHA — **Geneva, Switzerland**
Information Management Intern — *Jun 2026–Present*

- Built an **automated AI report-generation system** (not a chatbot): it monitors news, social and API sources and builds a fact and relationship graph of events, people and organisations (NLP, graph and vector databases, embeddings), so every statement is traceable to a named source.
- Containerised the UI with **Docker** for deployment on **Azure**; trained country offices to use and extend the system and wrote the accompanying guides.
- Contributed to Flash Appeal and Situation Analysis products and needs analysis for the **Venezuela 2026 earthquake**, mapping populated areas to surface overlooked populations under daily deadlines.

World Bank / CEU — **Vienna, Austria**
Researcher & Workstream Lead – Educational Inequality (TIMSS) — *Sep 2025–May 2026*

- Led the CEU team on a multi-country World Bank project (task allocation and milestones in Microsoft Planner) and was the interface with World Bank counterparts on deliverables.
- Cleaned and merged large-scale TIMSS data; ran **survey-weighted regression** and comparative regional analysis of socio-economic gaps with a reproducible methodology.

Independent **Bogotá, Colombia**
Independent Consultant – Data Science & GIS *Jul 2024–Jul 2025*

- Built ETL pipelines and **time-series models** (ARIMA, XGBoost, NARX) for air-quality and climate forecasting on Copernicus/Sentinel data; deployed dashboards in Dash, Plotly and Power BI.
- Created GeoJSON datasets for FM-radio coverage, environmental monitoring and infrastructure planning.

EU Delegation in Colombia **Bogotá, Colombia**
Administrative Assistant / Archivist *May 2024–Oct 2024*

- Designed and executed a five-month plan over 20 years of EU cooperation records (about 8 tons, 9,300 files); built Excel tracking that forecast progress with $\sim$**90% accuracy** and met 100% of the target.

Organization of American States (OAS) **Santo Domingo & El Salvador**
National Office Intern (Research) & International Electoral Observer *Jun 2023–Feb 2024*

- Analysed migration discourse by scraping presidential speeches and social-media data with **Python NLP**; drafted analytical reports and ran a digital strategy that lifted engagement by **500%+**.
- Observed the 2024 Salvadoran elections (DECO mission), documenting practices with a legal and gender focus.

I2T Laboratory & SegLab, Universidad ICESI **Cali, Colombia**
Research Assistant / Data Analyst *Feb 2021–Jul 2024*

- Applied ML and deep learning to geospatial health data linking environmental exposure to Parkinson's disease progression: a statistically significant association with proximity to crops, with no change in the gradient of Hoehn & Yahr stage or other disease measures (unpublished); presented to doctors, engineers and designers.
- Security and peacebuilding research with QGIS and Power BI dashboards feeding policy papers and articles.

EDUCATION

Central European University (CEU) **Vienna, Austria**
MA in Public Policy – Quantitative Policy Analysis *Sep 2025–Jun 2026*
Thesis: *(I'm)Paz Total: Quantitative Evaluation of the Política de Paz Total in Colombia*. Coursework: **Text Analysis & NLP**, Impact Evaluation (R), Public Procurement Analytics, Data Visualization.

Universidad ICESI **Cali, Colombia**
*BA in Anthropology (*Antropólogo*), Magna Cum Laude* *Feb 2019–Feb 2024*
GPA 4.5/5.0, ranked 1st in class, Dean's List. Erasmus exchange, University of Turin (2022).

SELECTED PROJECTS (GITHUB.COM/DANIEL-NAVARRO-M)

- **Paz Total impact evaluation:** monthly panel from National Police and ACLED data for three conflict regions; two-way fixed effects and event studies in R.
- **Rail Baltica corruption-risk analytics:** contract-level Corruption Risk Index from red-flag indicators validated by logistic regression on EU tender data (with P. Biša).
- **Migration discourse NLP:** seeded LDA, DBSCAN and transformer sentiment on 43 presidential speeches and $\sim$8,800 Reddit sentences.

PUBLICATIONS, AWARDS & LANGUAGES

Publication: Milanese & Navarro (2025). *Beyond municipal boundaries: metropolitanization of electoral behavior in Cali using k-means clustering.* Revista Geographia (Brazil).

Awards & Certifications: 1st Place, 2nd TechPolítica hackathon, FLACSO Ecuador • MEAL (Monitoring, Evaluation, Accountability & Learning) • Satellite Image Processing • Effective Project Planning & Evaluation in Health Research.

Languages: Spanish (native) • English (C1, IELTS Academic 8.0) • Italian (B1) • French (A2) • German (beginner).